System Coding Club

Reverse Engineering Challenge

Track C: SPICE Model Characterization

Welcome to Track C! In this track, your engineering intuition and analytical reasoning will be put to the test. You are provided with "black-box" equivalent block models. Your objective is to observe their input-output characteristics and reconstruct a functionally equivalent circuit.

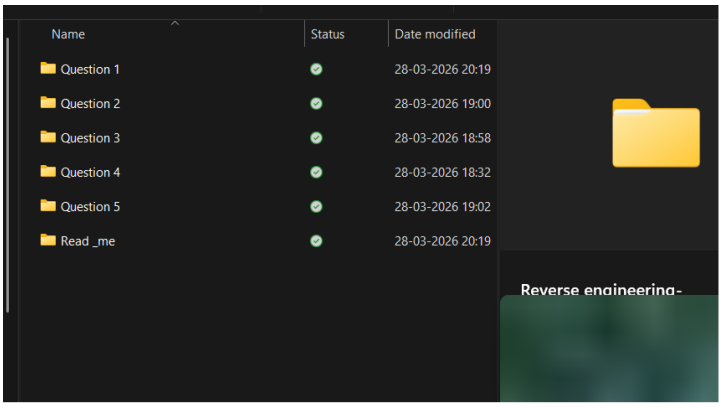
Please read the following instructions carefully to set up your LTspice environment and begin the challenge.

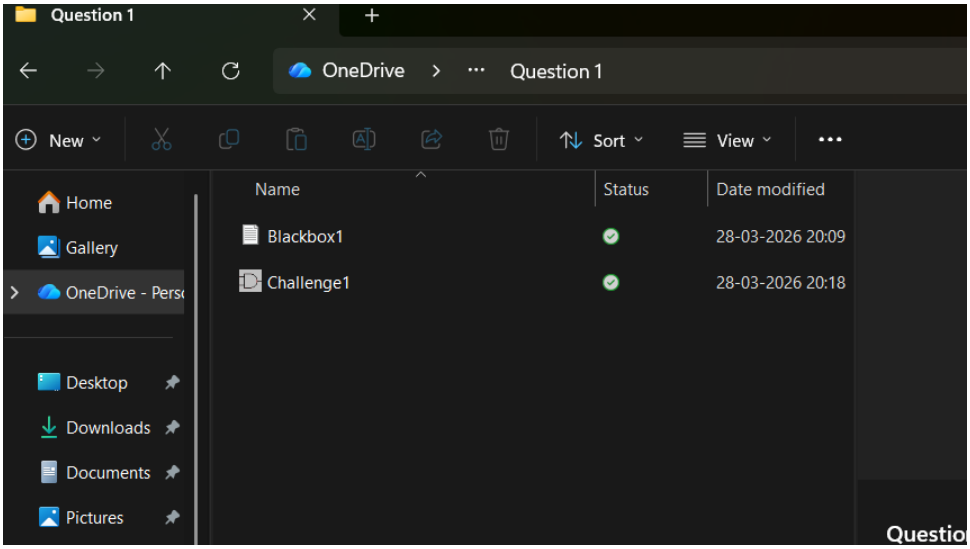
Step-by-Step Setup Guide

1. **Extract the Files:**
   First, completely unzip the provided challenge package to a local directory on your machine. Do not attempt to run the files directly from the zipped folder.

2. **Organize by Question:**
   Tackle the challenge question by question. Navigate into the specific folder for the question you are currently attempting (e.g., `Q1`, `Q2`).

3. **Create a New Schematic:**
   Open LTspice. Go to **File ¿ New Schematic** (or press `Ctrl+N`). Save this new schematic (`.asc` file) directly inside the specific question folder you are working on.


Name
Status
Date modified
Question 1
28-03-2026 20:19
Question 2
28-03-2026 19:00
Question 3
28-03-2026 18:58
Question 4
28-03-2026 18:32
Question 5
28-03-2026 19:02
Read _me
28-03-2026 20:19
Reverse engineering-


Question 1
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Question 1
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Blackbox1
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Challenge1
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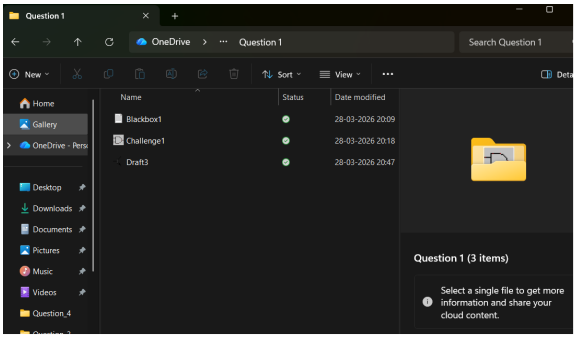
Questio


Question 1
OneDrive
Question 1
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Question_4
Name
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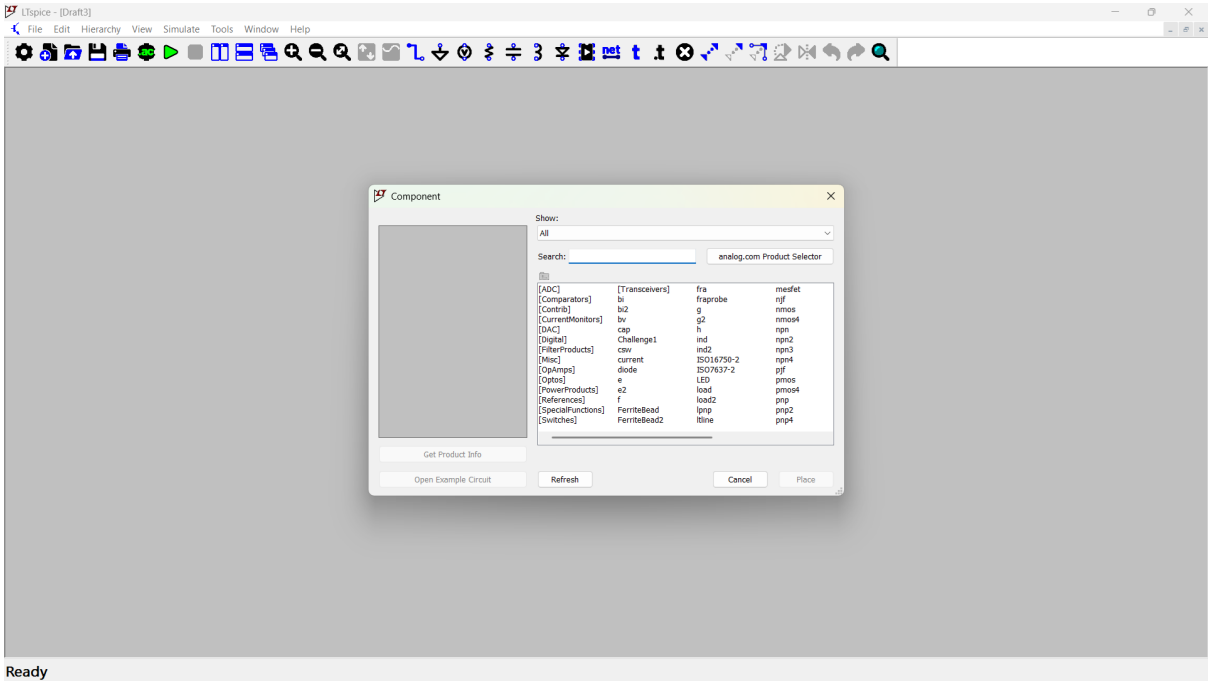
Date modified
Blackbox1
28-03-2026 20:09
Challenge1
28-03-2026 20:18
Draft3
28-03-2026 20:47
Question 1 (3 items)
Select a single file to get more information and share your cloud content.


LTspice - [Draft3]
File Edit Hierarchy View Simulate Tools Window Help
Component
Show:
All
Search:
analog.com Product Selector
[ADC]
[Comparators]
[Contrib]
[CurrentMonitors]
[DAC]
[Digital]
[FilterProducts]
[Misc]
[OpAmps]
[Optos]
[PowerProducts]
[References]
[SpecialFunctions]
[Switches]
[Transceivers]
bi
bi2
bv
cap
Challenge1
csw
current
diode
e
e2
f
FerriteBead
FerriteBead2
fra
fraprobe
g
g2
h
ind
ind2
ISO16750-2
ISO7637-2
LED
load
load2
lpnp
ltline
mesfet
njf
nmos
nmos4
npn
npn2
npn3
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pnp
pnp2
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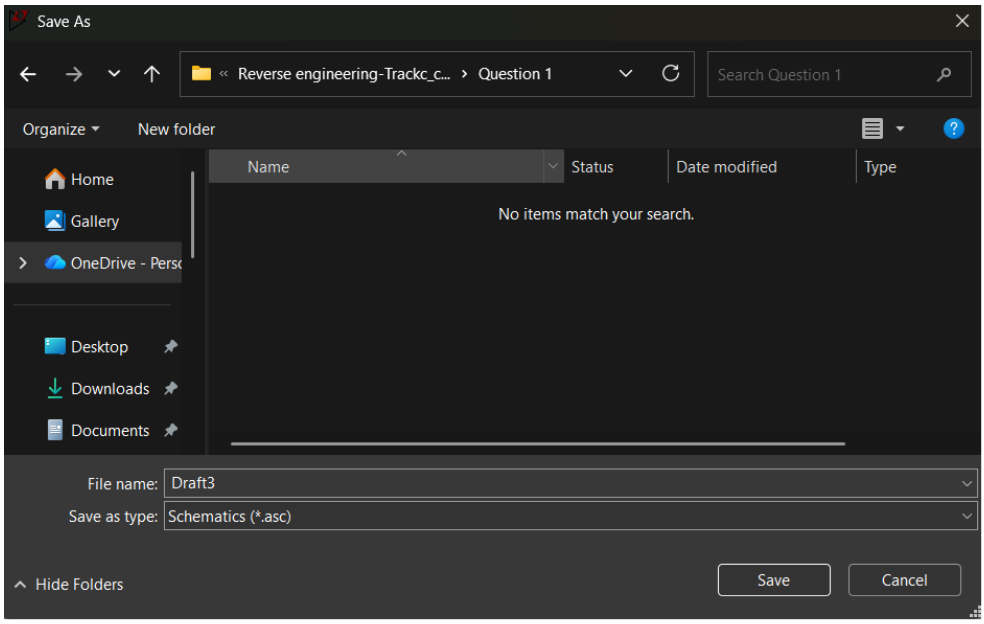
Get Product Info
Open Example Circuit
Refresh
Cancel
Place
Ready


Save As
Reverse engineering-Trackc_c... › Question 1
Search Question 1
Organize
New folder
Home
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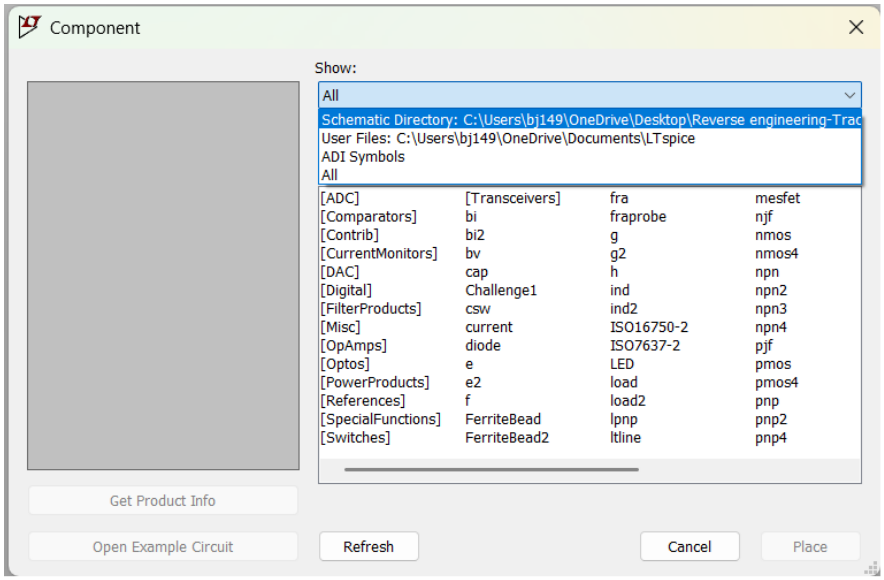
Documents
Name
Status
Date modified
Type
No items match your search.
File name: Draft3
Save as type: Schematics (*.asc)
Hide Folders
Save
Cancel


Component
Show:
All
Schematic Directory: C:\Users\bj149\OneDrive\Desktop\Reverse engineering-Trac
User Files: C:\Users\bj149\OneDrive\Documents\LTspice
ADI Symbols
All
[ADC]
[Comparators]
[Contrib]
[CurrentMonitors]
[DAC]
[Digital]
[FilterProducts]
[Misc]
[OpAmps]
[Optos]
[PowerProducts]
[References]
[SpecialFunctions]
[Switches]
[Transceivers]
bi
bi2
bv
cap
Challenge1
csw
current
diode
e
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f
FerriteBead
FerriteBead2
fra
fraprobe
g
g2
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ind
ind2
ISO16750-2
ISO7637-2
LED
load
load2
lpnp
ltline
mesfet
njf
nmos
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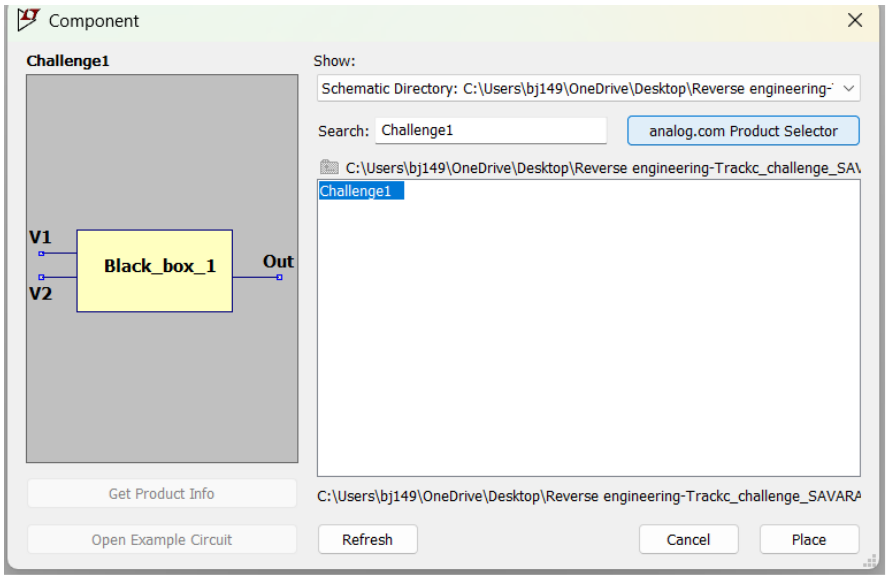
Get Product Info
Open Example Circuit
Refresh
Cancel
Place


Component
Challenge1
V1
V2
Black_box_1
Out
Show:
Schematic Directory: C:\Users\bj149\OneDrive\Desktop\Reverse engineering-
Search: Challenge1
analog.com Product Selector
C:\Users\bj149\OneDrive\Desktop\Reverse engineering-Trackc_challenge_SAV
Challenge1
Get Product Info
C:\Users\bj149\OneDrive\Desktop\Reverse engineering-Trackc_challenge_SAVARA
Open Example Circuit
Refresh
Cancel
Place

4. **Place the Black-Box Component:**
   To load the hidden system design into your schematic:

   - Click the **Component** button in the top toolbar (or press `F2`).
   - In the "Select Component Symbol" window, look at the **Top Directory** drop-down menu.
   - Change the directory to the specific question folder where you saved your schematic.
   - You should now see the custom component symbol (`.asy` file) for that question. Select it and click **OK** to place the block on your schematic.

5. **Add Required SPICE Directives:**
   You must link the underlying encrypted/protected subcircuit and standard libraries to the symbol.

   - Press `S` or `.` on your keyboard to open the **Edit Text on the Schematic** window.
   - Ensure the **"SPICE directive"** radio button is selected.
   - Add the following lines exactly as shown (make sure to replace `<YourUsername>` with your actual Windows username):

```
.model D D
.lib ADI.lib
.backanno
.lib C:\Users\<YourUsername>\AppData\Local\LTspice\lib\cmp\
   standard.dio
.lib C:\Users\<YourUsername>\AppData\Local\LTspice\lib\cmp\
   standard.mos
```

   - Place the text anywhere on your schematic.

6. **Apply Inputs:**
   Wire up your circuit. Connect the appropriate Voltage (`voltage`) or Current (`current`) sources to the input pins of the black-box component based on what you are trying to test.

7. **Simulate and Observe:**
   Set up your simulation commands to analyze the behavior:

   - **Time-Domain Response:** Use a Transient Analysis (`.tran`) to observe how the circuit responds to steps, pulses, or sine waves over time.
   - **Frequency Response:** Use an AC Analysis (`.ac`) to observe the bandwidth, gain, and phase shifts (Bode plot).
   - Run the simulation and probe the output nodes.

8. **Deduce and Reconstruct:**
   Analyze the generated graphs. Based on the time and frequency responses, deduce the internal circuitry. Create a *new* schematic mimicking this behavior using standard components.

9. **Submit Your Solution:**
   Once you have successfully reconstructed the circuits, zip your entire working folder. This zipped file should include your newly created `.asc` schematics and simulation result screenshots. Upload this zipped file to Google Drive (ensure link sharing is set to "Anyone with the link") or host it on a GitHub repository, and submit the link to the organizers.

Component Hints

While reconstructing the circuits, you may find the following components highly relevant to achieving behavioral equivalence:

- **Op-Amp:** `747` (LM747)
- **Diode:** `1N5817` (Schottky Diode)

Note: *Remember that your final score is based on the accuracy of the functional I/O characteristics, not on achieving an exact 1:1 structural duplicate of the hidden schematic.*

Good luck!